REFONTE INTERNATIONAL TRAINING AND INTERNSHIP PROGRAM

Astrodynamics Specialist

SESSION 7

Rendezvous and Proximity Operations

Relative Motion, the Clohessy-Wiltshire Equations, and How Spacecraft Dock

Prepared for	Cao Khanh Tan
Module	7, closing the Advanced Application Phase
Builds on	Sessions 1 to 3: orbital motion, vis-viva, impulsive delta-v
Tools	STK / GMAT relative-motion and rendezvous tools, Python

SESSION 7

Rendezvous and Proximity Operations

So far every orbit has been described from the centre of the Earth. But when one spacecraft has to reach another, dock with a station, capture a satellite for servicing, or inspect a piece of debris, what matters is the motion of the chaser relative to the target. In a frame that rides along with the target, that relative motion follows a strikingly simple set of linear equations, the Clohessy-Wiltshire equations, and they behave in ways that are famously counterintuitive. This module is where orbital mechanics becomes proximity operations, the last building block before the capstone.

Learning objectives

- Set up the local target-centred (Hill / LVLH) frame: radial, along-track and cross-track.
- Write the Clohessy-Wiltshire equations and state their assumptions.
- Use the closed-form CW solution to propagate relative motion.
- Explain why a radial offset causes an along-track drift, and state the no-drift condition.
- Solve the two-impulse rendezvous problem for the required burns.
- Compare V-bar and R-bar approaches and reason about passive safety.

> **The counterintuitive heart of it**
> In orbit you cannot simply thrust toward your target and close the gap. A push changes your orbit, and orbital mechanics curves your path. Rendezvous is the art of using those curved relative trajectories on purpose, and the Clohessy-Wiltshire equations are the map.

PART 1

The Target-Centred Frame

We place the origin on the target, which flies a circular reference orbit, and let the frame rotate with it. The three axes have names used throughout spaceflight operations: x is radial (the R-bar, pointing away from Earth), y is along-track (the V-bar, along the velocity), and z is cross-track (out of the orbit plane).

For a chaser close to the target, the relative equations of motion linearise into the Clohessy-Wiltshire (Hill) equations, where n is the mean motion of the reference orbit and the right-hand side is any thrust acceleration:

$$\ddot{x} - 2n\dot{y} - 3n^2x = a_x$$

$$\ddot{y} + 2n\dot{x} = a_y$$

$$\ddot{z} + n^2z = a_z$$

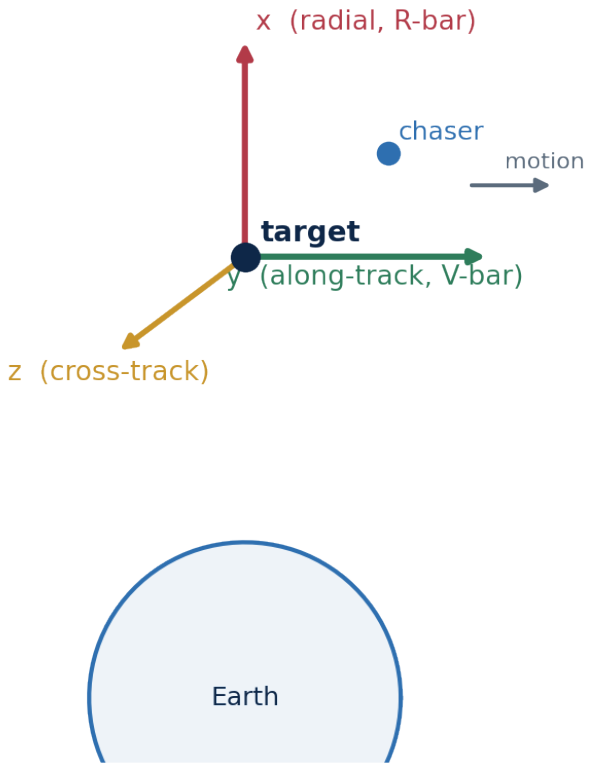


Figure 1. The Hill (LVLH) frame, centred on the target. The chaser position is measured in radial (R-bar), along-track (V-bar) and cross-track components. Approaches are named after the axis they come in along.

They assume a circular reference orbit, a separation small compared with the orbit radius, and two-body gravity. Notice the third equation stands alone: the cross-track motion is completely decoupled from the in-plane motion.

PART 2

The Closed-Form Solution and Natural Motion

The CW equations solve in closed form. Given the initial relative position and velocity, the position at any later time is:

$$x(t) = (4 - 3\cos nt)\,x_0 + \frac{\sin nt}{n}\,\dot{x}_0 + \frac{2}{n}(1 - \cos nt)\,\dot{y}_0$$

$$y(t) = 6(\sin nt - nt)\,x_0 + y_0 - \frac{2}{n}(1 - \cos nt)\,\dot{x}_0 + \frac{1}{n}(4\sin nt - 3nt)\,\dot{y}_0$$

$$z(t) = z_0 \cos nt + \frac{\dot{z}_0}{n}\sin nt$$

The cross-track equation is a simple harmonic oscillator: out-of-plane, the chaser just bobs up and down with the orbital period, never drifting. The in-plane motion is where the surprises live. A chaser sitting a little below the target, with no relative velocity, does not stay put: it swings low and drifts steadily ahead, because a lower orbit moves faster. A one kilometre radial offset drifts almost 38 km along-track in a single orbit.

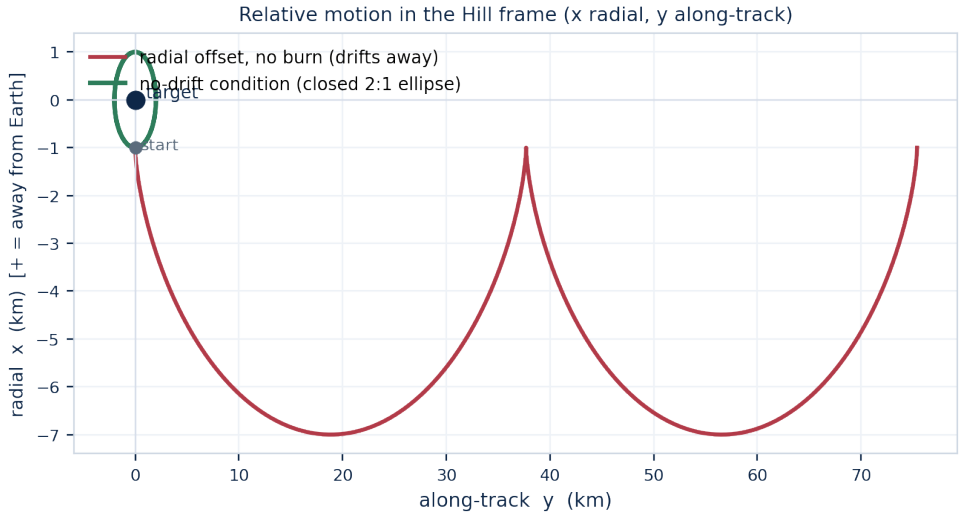


Figure 2. Natural relative motion. A pure radial offset with no burn (red) swings deep and drifts away along-track. Add just the right along-track velocity and the motion closes into a bounded 2:1 ellipse (green) that circles the target forever.

That bounded orbit is the key to safe proximity operations. The drifting terms in the solution vanish when the initial along-track velocity is tuned to the radial offset, the no-drift condition:

$$\dot{y}_0 = -2\,n\,x_0$$

When it holds, the relative trajectory is a closed ellipse whose along-track size is exactly twice its radial size, the natural motion circumnavigation used for inspection fly-arounds.

PART 3

Two-Impulse Rendezvous

To actually reach the target, the chaser fires two burns. The first sets it onto a transfer trajectory that arrives at the target after a chosen time; the second cancels the closing velocity so it arrives at rest. Because the CW solution is linear, the required starting velocity comes straight from inverting the position relation:

$$\mathbf{v}_0^{+} = \Phi_{rv}^{-1}(T)\left[\mathbf{r}_f - \Phi_{rr}(T)\,\mathbf{r}_0\right]$$

where the Phi matrices are the blocks of the CW state-transition matrix over the transfer time T. The first burn is the difference between that required velocity and the current one; the second nulls the arrival velocity. The total cost is:

$$\Delta v = \left|\mathbf{v}_0^{+} - \mathbf{v}_0^{-}\right| + \left|\mathbf{v}_f\right|$$

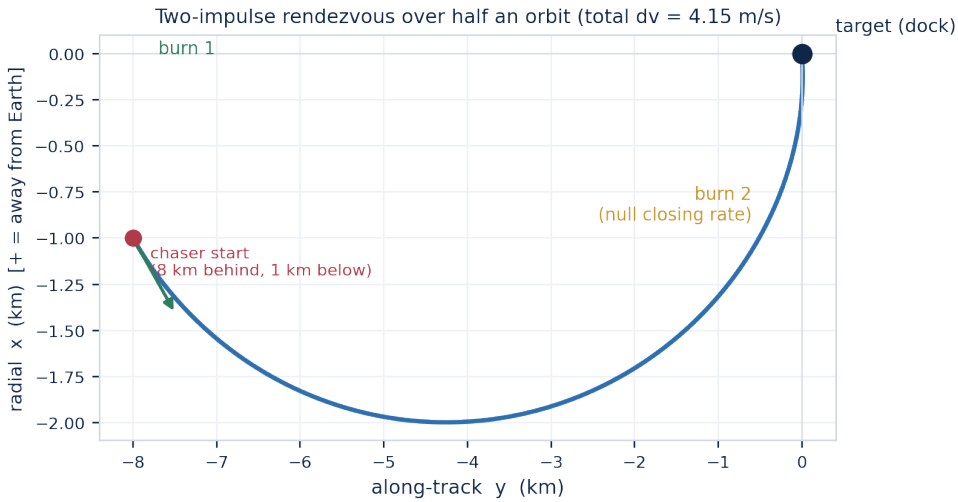


Figure 3. A two-impulse rendezvous. The chaser starts 8 km behind and 1 km below, burns onto a half-orbit transfer, coasts along the curved relative path, and brakes to a stop at the target.

Worked example: an 8 km rendezvous

A chaser 8 km behind and 1 km below the target, initially drifting with it, wants to dock in half an orbit (about 46 minutes at ISS altitude). Solving the CW targeting problem gives a first burn of 2.53 m/s and a braking burn of 1.61 m/s, a total of 4.15 m/s. Note the transfer time cannot be exactly one full orbit, where the targeting matrix becomes singular, a real constraint in mission planning.

PART 4

V-bar, R-bar and Passive Safety

The final approach is flown along one of the frame axes, and the choice matters for safety as much as fuel:

- **V-bar approach (along-track):** the chaser closes from ahead of or behind the target, along the velocity vector. It is intuitive but not passively safe: if a braking burn is missed, CW dynamics carry the chaser toward the target.
- **R-bar approach (radial):** the chaser rises or descends toward the target along the local vertical. Orbital mechanics naturally slows the approach, so a missed burn causes the chaser to drift away rather than collide. This passive safety is why cargo vehicles berth with the ISS from below on the R-bar.

Real proximity operations chain these together: a far approach on the V-bar to a hold point, a switch to an R-bar corridor for the final few hundred metres, and a slow closing rate under continuous monitoring. Every leg is a CW transfer, and every hold point is a bounded relative orbit.

Computing Relative Motion

The CW state-transition matrix is a few lines, and it both propagates relative motion and solves the two-impulse targeting problem by inversion.

```
import numpy as np
mu = 398600.4418; a = 6378.137 + 420; n = np.sqrt(mu/a**3)

def cw(t):
    s, c = np.sin(n*t), np.cos(n*t)
    Prr = np.array([[4-3*c,0,0],[6*(s-n*t),1,0],[0,0,c]])
    Prv = np.array([[s/n,2*(1-c)/n,0],[-2*(1-c)/n,(4*s-3*n*t)/n,0],[0,0,s/n]])
    Pvr = np.array([[3*n*s,0,0],[6*n*(c-1),0,0],[0,0,-n*s]])
    Pvv = np.array([[c,2*s,0],[-2*s,4*c-3,0],[0,0,c]])
    return Prr, Prv, Pvr, Pvv

r0 = np.array([-1.0,-8.0,0.0]); T = np.pi/n        # half orbit
Prr,Prv,Pvr,Pvv = cw(T)
v0 = np.linalg.solve(Prv, -Prr@r0)                 # burn 1 target velocity
vf = Pvr@r0 + Pvv@v0                                # arrival velocity
dv = np.linalg.norm(v0) + np.linalg.norm(vf)        # ~4.15e-3 km/s
```

Production tools

STK and GMAT both model relative motion directly: STK Astrogator has rendezvous and relative-state targeters, and its proximity tools plot the trajectory in the RIC (radial, in-track, cross-track) frame you see in the figures. For high fidelity they replace the linear CW model with full nonlinear propagation, but the CW result is the sanity check and the first guess every targeter starts from.

PRACTICE

This Week (aim for 12 to 15 hours)

Bring your work and plots to the next live session. As always, show your reasoning, not just the final numbers.

Assignment: Rendezvous and Proximity Operations

- **Reading and notes (3 to 4 h).** State the assumptions behind the CW equations and explain, in your own words, why the cross-track motion decouples and why a radial offset produces an along-track drift. One page.
- **Natural motion (3 to 4 h).** For a chaser 2 km below the target with no relative velocity, compute the along-track drift per orbit and the radial swing. Then find the along-track velocity that turns it into a bounded 2:1 ellipse (the no-drift condition).
- **Two-impulse rendezvous (4 to 5 h).** A chaser is 15 km behind and 2 km below the target. Using the CW state-transition matrix, compute the two burns and the total delta-v to dock in half an orbit. Plot the relative trajectory.
- **Cross-track (1 to 2 h).** A chaser has a 1 km cross-track offset and no cross-track velocity. Describe its motion and give its period.
- **STK or GMAT (2 h).** Set up a target and chaser in STK, view the relative motion in the RIC frame, and reproduce a bounded fly-around or a simple two-burn approach. Capture the relative-motion plot.

Answer key (for your self-check)

Quantity	Expected value
Cross-track oscillation period	one orbital period, about 93 min at ISS altitude
2 km radial offset: along-track drift	about 75 km per orbit; radial swings from +2 to -10 km
No-drift velocity for x0 = 2 km	ydot0 = -2 n x0, about -4.51 m/s
Two-impulse (2 km below, 15 km behind, half orbit)	dv1 about 4.89, dv2 about 2.95, total about 7.84 m/s

REFERENCE

One-Page Formula Sheet

Mean motion	$n = \sqrt{\mu / a^3}$
CW equation, radial	$\ddot{x} - 2n\dot{y} - 3n^2 x = a_x$
CW equation, along-track	$\ddot{y} + 2n\dot{x} = a_y$
CW equation, cross-track	$\ddot{z} + n^2 z = a_z$
Solution, radial	$x(t) = (4 - 3\cos nt)\,x_0 + \frac{\sin nt}{n}\,\dot{x}_0 + \frac{2}{n}(1 - \cos nt)\,\dot{y}_0$
Solution, along-track	$y(t) = 6(\sin nt - nt)\,x_0 + y_0 - \frac{2}{n}(1 - \cos nt)\,\dot{x}_0 + \frac{1}{n}(4\sin nt - 3nt)\,\dot{y}_0$
Solution, cross-track	$z(t) = z_0 \cos nt + \frac{\dot{z}_0}{n} \sin nt$
No-drift (bounded orbit)	$\dot{y}_0 = -2\,n\,x_0$
Two-impulse targeting	$\mathbf{v}_0^+ = \Phi_{rv}^{-1}(T)[\mathbf{r}_f - \Phi_{rr}(T)\,\mathbf{r}_0]$
Rendezvous total delta-v	$\Delta v = \lvert\mathbf{v}_0^+ - \mathbf{v}_0^-\rvert + \lvert\mathbf{v}_f\rvert$

Key glossary

- **Hill / LVLH frame:** the rotating, target-centred frame of radial, along-track and cross-track axes.
- **R-bar / V-bar:** the radial and along-track directions; approaches are named after them.
- **No-drift condition:** the along-track velocity (ydot0 = -2 n x0) that makes relative motion a closed ellipse.
- **Two-impulse rendezvous:** a burn onto a transfer trajectory and a braking burn at arrival.
- **Passive safety:** an approach (R-bar) where a missed burn drifts the chaser clear rather than into the target.

Going further

For depth, see Curtis, Orbital Mechanics for Engineering Students (relative motion and rendezvous), Vallado, Fundamentals of Astrodynamics and Applications (the CW model and targeting), and Fehse, Automated Rendezvous and Docking of Spacecraft, for the operational side. These are real, widely used texts; please confirm editions yourself, as I cannot verify those details from here.

End of the taught modules: coming up, the Capstone

Sessions 1 to 7 have built the full toolkit: two-body motion, perturbations, transfers, interplanetary trajectories, constellations, station-keeping and rendezvous. Session 8 is the capstone: a complete mission of your choice (an Earth-to-Mars transfer, a GNSS constellation design, or a station-keeping optimisation) with a full report and oral defence.